SOFTWARE TRAINING REPORT · VIVA PRESENTATION

Customer Churn Prediction Using Machine Learning

A Machine Learning Approach to Predicting & Preventing Customer Loss

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CLASS / SECTION

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The Problem

Why customer churn matters to every subscription business

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more expensive to acquire a new customer than to retain an existing one

“Can we predict, in advance, which customers are at high risk of leaving?”

— the Key Question this project answers

Rich Historical Data

Tenure, monthly charges, contract type and service usage already exist for every customer.

Early Warning Signals

Patterns in this data reveal which customers are quietly drifting toward cancellation.

Business Impact

Flagging at-risk customers early lets a company step in with retention offers before it's too late.

Objectives & Scope

What the project sets out to do, and where it applies

OBJECTIVES

- Analyse customer data to find key churn factors
- Build & train a Logistic Regression classifier
- Evaluate using accuracy, confusion matrix & report
- Build a real-time prediction module

SCOPE

- Structured, tabular customer dataset — ID, gender, age, tenure, charges, contract, internet service
- Binary classification (churn vs. no-churn) using supervised learning
- Extendable to any subscription business — telecom, banking, OTT, insurance, gyms

Dataset Overview

A structured, tabular customer record used to train the model

Feature	Description
CustomerID	Unique customer identifier
Gender / Age	Demographic details of the customer
Tenure	Number of months with the company
Monthly / Total Charges	Monthly payment and total amount paid
Contract	Contract type — temporary or permanent
Internet Service	Type of internet service used
Churn (Target)	0 = Customer stays, 1 = Customer leaves

Methodology

Preprocessing → Model Building → Testing & Evaluation

1

Analyse & Clean

Study dataset shape/structure; fill missing values with column mean

2

Encode & Scale

Label-encode categorical fields; StandardScaler brings all features to one scale

3

Split 80/20

80% training data, 20% held-out test data

4

Train Model

Logistic Regression — sigmoid outputs a probability between 0 and 1

5

Predict & Evaluate

Test on unseen data; score with accuracy, confusion matrix, classification report

Tools & Technologies

The stack used to build, train and evaluate the model

PLATFORM

Python 3.x on Google Colab / Jupyter Notebook, Windows 10/11

CORE LIBRARIES

NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, Pickle

HARDWARE

Intel i3 or higher · 4 GB RAM or higher · 20 GB free disk space

Results & Evaluation

How the trained model was scored on unseen test data

CONFUSION MATRIX

	Pred: Stay	Pred: Leave
Actual: Stay	TN	FP
Actual: Leave	FN	TP

Reported via scikit-learn's classification_report() on the held-out 20% test set.

Accuracy

Overall % of correct predictions

Precision

Of predicted churners, how many actually churned

Recall

Of actual churners, how many were correctly caught

F1-Score

Balance between Precision and Recall

Value Addition & Limitations

What the project delivers, and where it currently falls short

VALUE ADDITION

- Enables proactive identification of at-risk customers instead of reacting after they leave
- Helps design targeted retention offers, potentially saving revenue
- Provides a reusable, saved (pickled) model + scaler ready for deployment
- Serves as a reusable ML pipeline template for other classification problems

LIMITATIONS

- Trained on a limited, static dataset (~1000 records)
- Only Logistic Regression used — no comparison with other algorithms
- Works on an offline dataset, not connected to a live CRM/database
- Label Encoding may not fully capture nominal category relationships

Conclusion & Future Scope

A complete ML pipeline — cleaning, encoding, scaling, training and evaluation — giving businesses an actionable early-warning tool for customer retention.

Compare Algorithms

Random Forest, Decision Tree, SVM, XGBoost

Tune & Validate

Hyperparameter tuning & cross-validation for robustness

Deploy as Web App

Flask / Streamlit with a simple user interface

Live Integration

Connect to a continuously-updating company database

Richer Features

One-Hot Encoding & richer features for better accuracy

Thank You